

Trainings ▾

Disassemble

⚠ WARNING ⚠

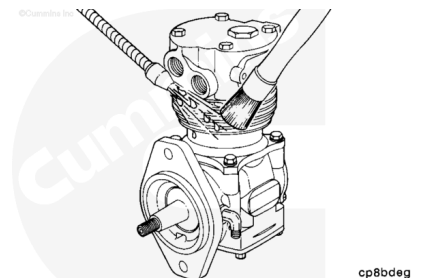
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

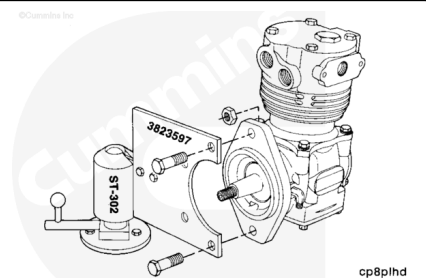
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam to clean the air compressor.

Dry the air compressor with compressed air.



Mount the compressor to the mounting plate, Part Number ST-749, or equivalent, which is used with the ball joint vise, Part Number ST-302.



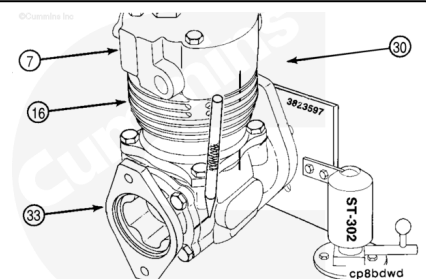
Before disassembling the air compressor, scribe the following parts to help alignment during assembly.

Cylinder head (7).

Cylinder block (16).

Crankcase (30).

Power steering adapter (33).

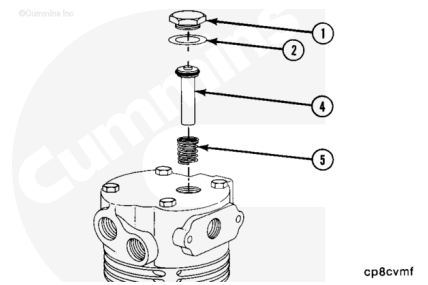


Remove the following parts:

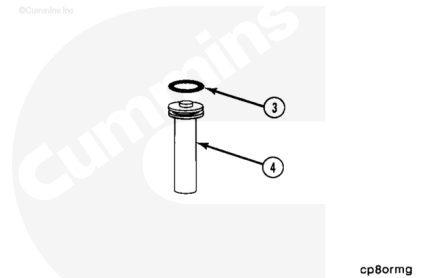
- Unloader cover (1)
- Copper washer (2)

- Unloader pin (4)
- Spring (5).

Discard the copper washer.



Remove and discard the o-ring (3) from the unloader pin (4).

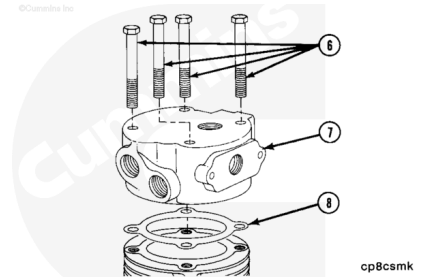


Note : Service Tip: Scribe a mark to show proper head orientation before removing the head.

Remove the four cylinder head capscrews (6).

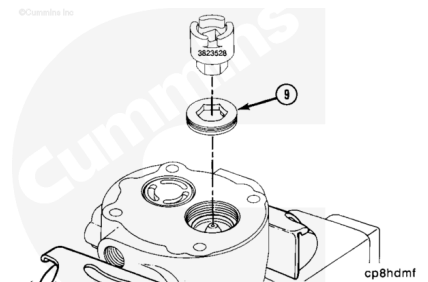
Remove the cylinder head (7).

Remove and discard the cylinder head gasket (8).



Install the head with the bottom side up in a soft-jawed vise.

Use an air compressor seat socket, Part Number 3823528, or equivalent, or a 19 mm [3/4 in] allen wrench, or equivalent, to remove the exhaust valve seat (9).



Note : The exhaust valve stop is pressed in place and must not be removed.

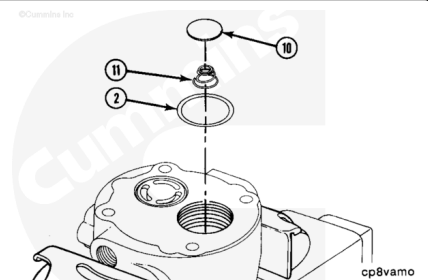
Remove the following parts:

Exhaust valve disc (10)

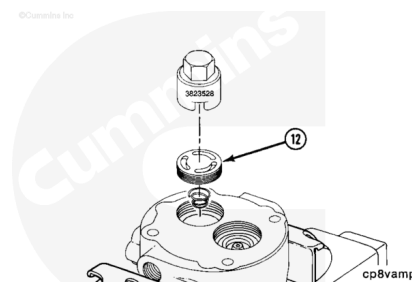
Spring (11)

Copper washer (2).

Discard the copper washer.



Use an air compressor seat socket, Part Number 3823528, or equivalent, to remove the inlet valve cage (12).



Remove the following parts:

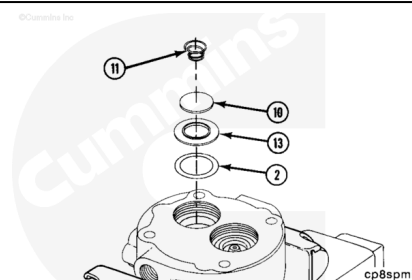
Spring (11).

Inlet valve disc (10).

Inlet valve seat (13).

Copper washer (2).

Discard the copper washer.



Clean and Inspect for Reuse

⚠ WARNING ⚠

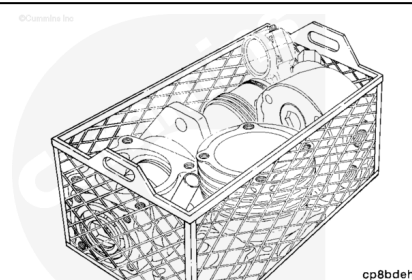
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, and soak time.

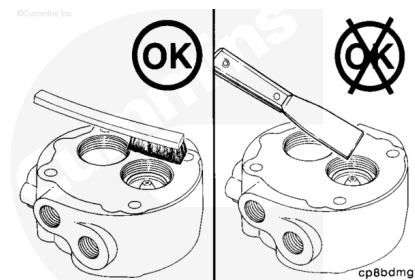
Dry with compressed air.



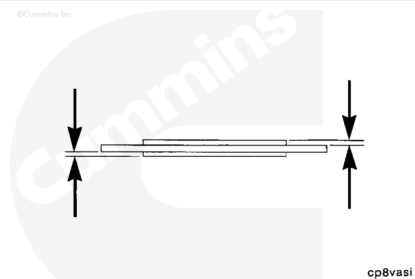
⚠ CAUTION ⚠

Do not use a scraper to remove carbon and scale. This can damage sealing surfaces.

Use a stiff, nonmetallic bristle brush to scrub the parts.

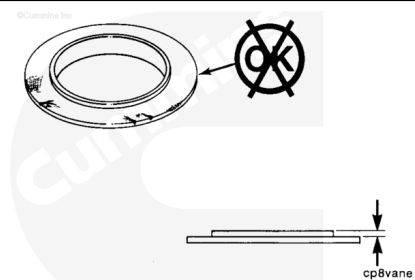


Inspect the valve discs and replace if cracked, pitted, or grooved in excess of 0.13 mm [0.005 in].



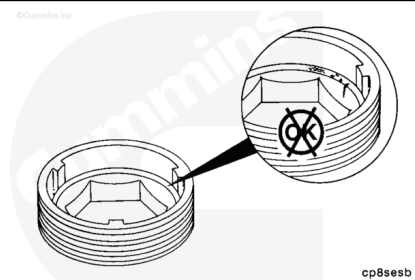
Measure the distance from the valve seating surface to the surface that contacts the valve cage.

mm		in
0.597	MIN	0.0235
0.673	MAX	0.0265



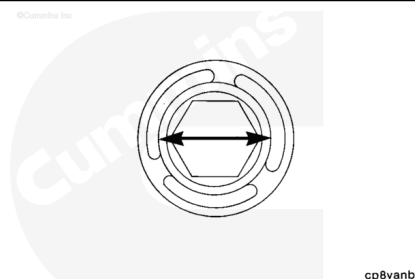
Replace the intake valve seat if **not** within limits or if cracked or damaged.

Inspect the exhaust valve seat for damage and wear.



Measure the valve guide diameter.

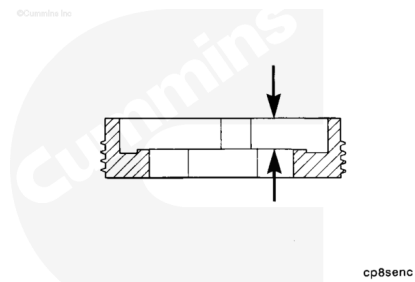
mm		in
25.53	MIN	1.005
25.65	MAX	1.010



Replace the exhaust valve seat if **not** within specification.

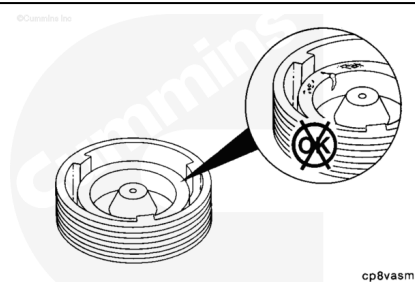
Measure the distance from the top of the valve seat to the valve seating surface.

mm		in
4.01	MIN	0.158
4.11	MAX	0.162



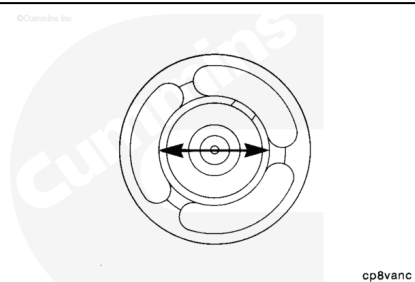
Replace the exhaust valve seat if **not** within limits.

Inspect the inlet valve cage for damage and wear.



Measure the valve guide diameter.

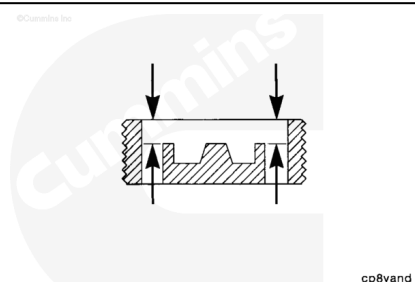
mm		in
25.53	MIN	1.005
25.65	MAX	1.010



Replace the inlet valve cage if **not** within limits.

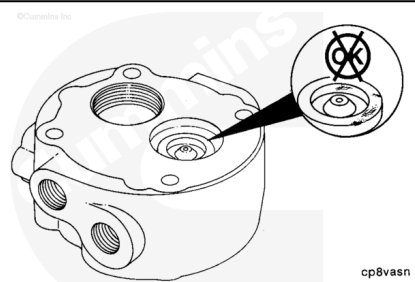
Measure the top of the cage to the valve stop.

mm		in
3.63	MIN	0.143
3.78	MAX	0.149



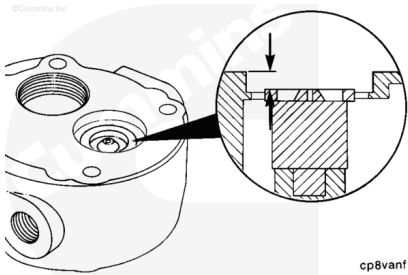
Inspect the exhaust valve stop.

Replace the cylinder head assembly, if the stop is loose or damaged.



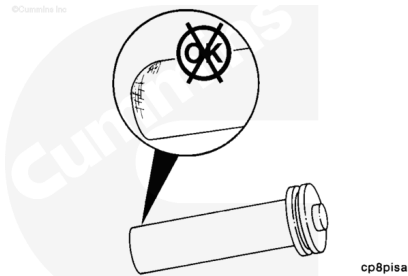
Measure the distance from the valve end of the stop to the face of the cylinder head.

mm		in
4.42	MIN	0.174
4.70	MAX	0.185



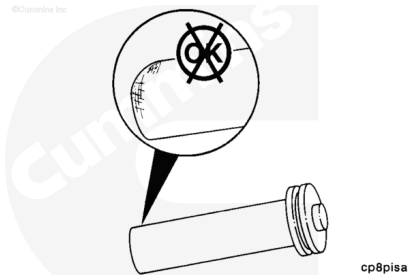
Replace the cylinder head if **not** within limits.

Inspect the unloader pin for scoring or pitting.



Measure the length of the pin.

mm		in
40.51	MIN	1.595
40.72	MAX	1.603

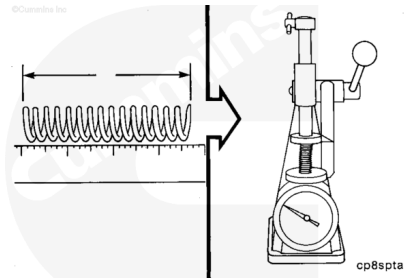


Replace the pin if **not** within limits.

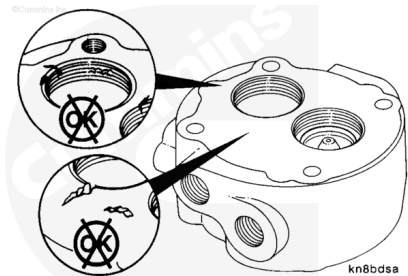
Note : Cummins Inc. recommends that new springs be installed during rebuild.

Use valve spring tester, Part Number 3375182, or equivalent, to check the springs.

Reference the Compressor Spring Force Specifications Chart in Section V.

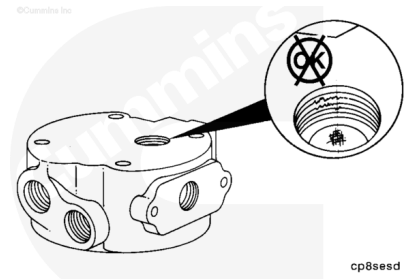


Inspect the air compressor cylinder head for cracks, nicks, gouges, or damaged threads. Replace, if necessary.



Inspect the unloader seal bore for scoring or pitting.

Replace the cylinder head, if damaged.



Assemble

Turn the cylinder head bottom side up, and install it in a soft-jawed vise.

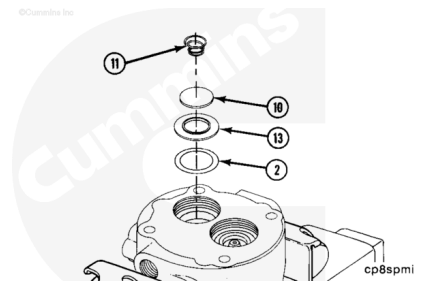
Install the following parts.

New washer (2).

Inlet valve seat (13).

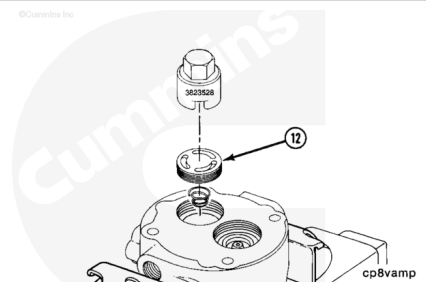
Inlet valve (10).

Inlet valve spring (11).



Use an air compressor seat socket, Part Number 3823528, or equivalent, to tighten the cage (12).

Torque Value: 108 n•m [80 ft-lb]

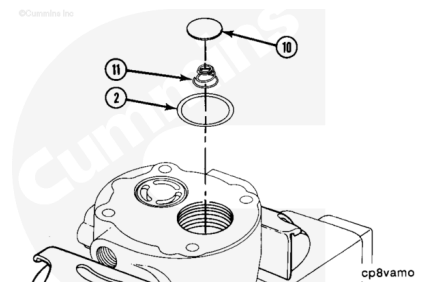


Install the following parts:

New washer (2).

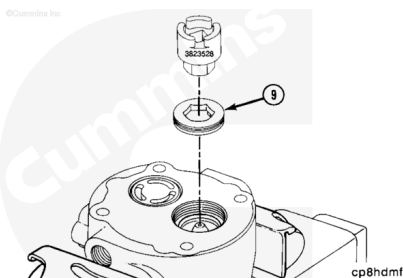
Exhaust valve spring (11).

Exhaust valve disc (10).



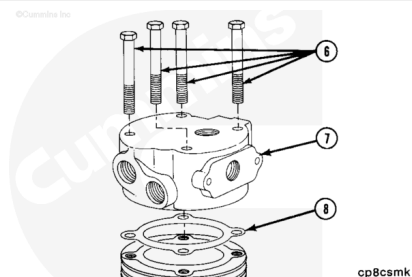
Use an air compressor seat socket, Part Number 3823528, or equivalent, or a 19 mm [3/4-in] allen wrench to tighten the seat (9).

Torque Value: 108 n•m [80 ft-lb]



Install a new gasket (8) and the cylinder head (7) to the cylinder block, aligning the scribe marks.

Install the four capscrews (6).

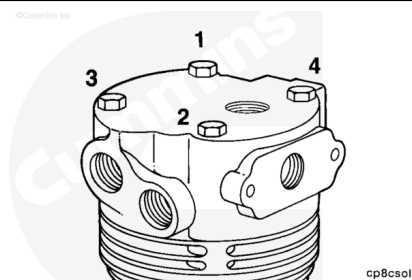


Tighten the capscrews.

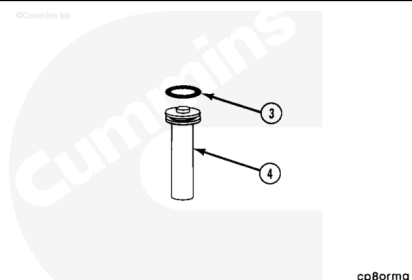
Torque Value: 30 n•m [22 ft-lb]

Tighten the capscrews again, in the sequence shown.

Torque Value: 41 n•m [30 ft-lb]



Install a new o-ring (3) on the unloader pin (4).



Install the following parts:

Spring (5)

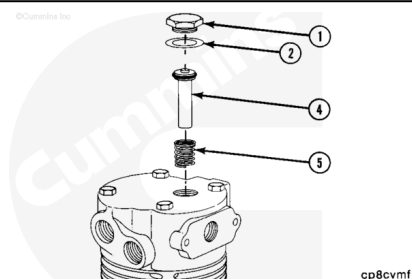
Unloader pin (4)

New washer (2)

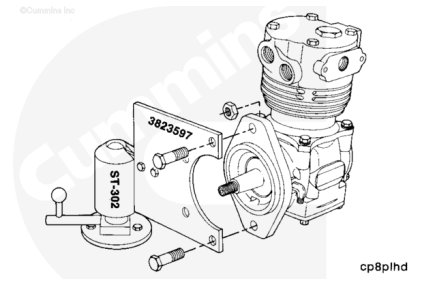
Unloader cover (1).

Tighten the cover.

Torque Value: 41 n•m [30 ft-lb]



Remove the air compressor from the mounting plate, Part Number ST-749, or equivalent.



Last Modified: 13-Jun-2005
